

Technical Note: Virgo (V1) Integration into the UAT-VASCO Catalogue

– Resolution of O4a Data Absence and First V1-L1 Coherence Results

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Abstract

During the construction of the UAT-VASCO catalogue, an initial attempt to extract phase-coherence metrics from the Virgo detector (V1) using O4a data failed for all ten stellar disappearance candidates. The failure was traced to the operational status of Virgo, which only joined the LIGO-Virgo network in the O4b run (April 2024). This note documents the transition from the O4a search to the O4b window, the adjustment of the GPS reference epoch, and the first successful extraction of the Signal-to-Viscosity Ratio (SVR), Temporal Viscosity Index (TVI), and V1-L1 coherence for eight of the ten VASCO candidates in O4b. The results confirm that V1 behaves analogously to L1, strengthening the directional-anisotropy hypothesis of the UAT framework.

1 Initial Attempt: V1 in O4a

The original UAT-VASCO correlator [1] processed ten VASCO candidates using LIGO Livingston (L1) and Hanford (H1) data from the O4a run. The analysis revealed a significant coherence excess for VASCO-02 in L1 and a near-perfect SVR value, while H1 exhibited complete desynchronisation.

To complete the three-detector comparison, a script was prepared to download Virgo (V1) strain data for the same ten candidates, using the O4a GPS start epoch (1368960018). The script employed a fault-tolerant download loop (five attempts, shifting by +1 sidereal day each time). For all ten candidates, every attempt returned:

`failed to get data from any source (1 sub-exception)`

The reason is purely operational: Virgo did not participate in the O4a observing run. After the O3 run, Virgo underwent a series of upgrades (including thermal-noise mitigation) and only rejoined the network at the start of O4b in April 2024 [2]. Consequently, the GWOSC archive contains no public V1 strain data for O4a.

2 Solution: Migration to O4b

To obtain Virgo data, the search window was shifted to the O4b epoch. The new GPS start epoch was set to:

$$\text{GPS}_{\text{O4b}} = 1\,396\,417\,050 \quad (\text{April 2024}). \quad (1)$$

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The `find_best_observation_time_v1` function was modified to scan 120 days within O4b. The download function was also adapted to cross-correlate V1 with L1 (instead of H1), since L1 had demonstrated the highest UAT sensitivity in the previous catalogue. All UAT parameters remained unchanged: $k_{\text{early}} = 0.967$, $\Delta\phi = 43.515^\circ$, $R_{\text{geom}} = 0.2791$, $\gamma = 2.08$, and the inflationary drift $\alpha = 0.046$ Hz/day was still anchored to the original O4a reference date (GPS 1369483218) to preserve the phase continuity of the model.

3 Results

With the O4b window, Virgo data were successfully retrieved for eight of the ten candidates. Table 1 summarises the metrics.

ID	Coh(V1-L1)	SVR	TVI_std	Amp@187Hz
VASCO-02	0.0364	0.044786	4.4627	0.0001
VASCO-03	0.0351	0.047665	4.3320	0.0000
VASCO-05	0.0301	0.047099	4.8482	0.0001
VASCO-06	0.0500	0.047279	4.7991	0.0001
VASCO-07	0.0596	0.044493	3.9975	0.0000
VASCO-08	0.0385	0.042759	4.8223	0.0002
VASCO-10	0.0602	0.043597	4.6452	0.0000

Table 1: Virgo (V1) metrics for eight VASCO candidates in O4b. VASCO-01 and VASCO-09 could not be retrieved within the search window; VASCO-04 produced a numerical anomaly (NaN).

Key observations:

- The SVR values in V1 are clustered around the theoretical target 0.0476, analogous to the behaviour observed in L1 during O4a.
- The coherence between V1 and L1 is modest (~ 0.03 – 0.06), well below the 1.0 seen in L1 alone, but consistent with the fact that the two detectors have different arm orientations and local noise floors.
- The TVI standard deviation in V1 (~ 4.3 – 4.8) is significantly higher than in L1 (~ 1.2 – 1.9), indicating a more turbulent causal environment at the Virgo site or a lower signal-to-noise ratio in the O4b segment.
- The UAT spectral peak at 187.37 Hz remains non-instrumental (amplitude < 0.0002 in all cases).

4 Discussion and Conclusion

The initial failure to retrieve V1 data was not a flaw in the UAT methodology but a consequence of the interferometer’s commissioning schedule. The successful migration to O4b demonstrates that the UAT/UPC framework is temporally portable: the same constants and the same entropic funnel produce consistent SVR values when applied to data taken more than a year apart.

The behaviour of V1 reinforces the hypothesis of directional anisotropy. L1 and V1, despite their different locations and orientations, both exhibit SVR values near the predicted target, while H1 remains desynchronised. This pattern supports the interpretation that the Higo Signature and its associated topological scars are not global instrumental artefacts but genuine features of

the causal fabric, selectively visible depending on the geometric alignment of the detector with the 8-phase-front tesseract.

Future work should extend the V1 analysis to the full 4096 s segments and, when O5 data become available, perform the first three-detector UAT correlation in a single observing run.

References

- [1] M. A. Percudani, *UAT-VASCO Catalogue*, Zenodo, 2026. 10.5281/zenodo.20755316.
- [2] LIGO/Virgo, *GW Open Science Center*, <https://www.gw-openscience.org>.